

GestureNet: A Robust Deep Learning Framework for Real time Sign Language Detection

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Abstract

Sign language detection is an important application of computer vision and deep learning, as it helps reduce the communication gap between the Deaf community and hearing community. Sign language recognition systems aim to interpret hand gesture inputs and express them in textual form and speech for effective communication. This paper presents a practical study on American Sign Language (ASL) recognition that highlights the challenges faced in quality of the data, in model selection and implementation in real time scenarios. Initially, a customized Convolutional Neural Network (CNN) was developed using gray scale ASL images of size 64×64 . The architecture consisted of three convolutional layers with ReLU activation, followed by max-pooling layers and a fully connected classification head for 36 sign classes. Even though the model with high training and validation accuracy, it performed poorly on externally captured images, indicating over fitting and weak generalization. Further analysis revealed that the dataset contained highly similar and duplicated images, limited background variation and potential data leakage between training and validation sets. To address these issues, extensive data preprocessing was applied, including duplicate image removal, class balancing through data augmentation and controlled dataset splitting. Transfer learning using MobileNetV2 was then employed to improve feature extraction and robustness. Experimental results demonstrate that the transfer learning-based model achieves better generalization and more reliable performance on unseen data. This work highlights the importance of proper dataset management and appropriate model selection for building practical sign language recognition systems.

Keywords: Sign Language Detection, American Sign Language, Convolutional Neural Network, Transfer Learning, MobileNetV2.

1 Introduction

Sign language represents a critical form of communication used by the Deaf or hard of hearing individuals and the hearing community. Automatic recognition of sign language using computer vision and deep learning techniques has gained significant attention, as it can help bridge the communication gap by enabling gesture-to-text and gesture-to-speech translation systems [1, 8 and 26]. Recent advances in deep learning-based image classification have significantly enhanced the recognition of hand gestures and sign language symbols by enabling automatic learning of discriminative visual features [3, 6 and 25].

American Sign Language (ASL) dataset is one of the most used sign language dataset, it contains alphabets (A-Z) and digits (0-9) are represented using distinct hand gestures. However, accurate recognition of ASL gestures remains challenging due to variations in hand shape, orientation, illumination conditions, background clutter and visual similarity between certain gestures [4, 13 and 19]. Furthermore, real-world sign language datasets often contain redundant, duplicated, or low-quality samples, which can negatively impact model generalization and reliability [22, 29].

This research focuses on the design of a sign language recognition system using deep neural networks. American Sign Language (ASL) image dataset containing hand gesture images of digits (0-9) and alphabets (A-Z). To improve dataset quality, extensive data preprocessing and cleaning were performed, including the removal of corrupted samples and elimination of duplicate images. Similar data quality issues, preprocessing strategies have been reported as critical steps in recent sign language recognition studies [21, 28]. After dataset cleaning, significant class imbalance was observed, with some gesture classes containing fewer samples than others. The problem was resolved by enriching the dataset through augmentation techniques like rotation, scaling, shifting, and flipping were applied to the training data to balance class distribution and improve model robustness, as commonly adopted in prior works [22, 27].

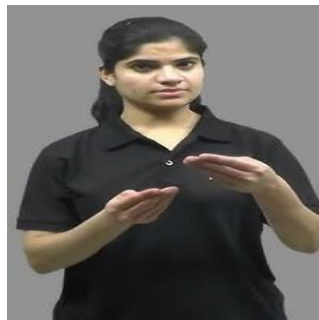


Fig. 1. Demonstration of Sign Language.

Multiple deep learning models were implemented and evaluated on the same dataset. Initially, a customized Convolutional Neural Network (CNN) and the InceptionV3 model were trained. Although both models achieved high training and validation accuracy, they suffered from over fitting and showed poor generalization on unseen real-world images, a limitation also observed in earlier sign language recognition studies [6, 18]. A detailed numerical comparison of these baseline models is presented in the Results section.

To overcome these limitations, MobileNetV2 was selected as the proposed model due to its lightweight architecture and strong feature extraction capability; this makes the model best suitable for resource-constrained and real-time applications [20]. The model was implemented using transfer learning with pre-trained weights of Image Net dataset, followed by selective fine-tuning of deeper layers to learn gesture-specific features while retaining robust low level representations [24, 25]. The findings validate that the proposed approach delivers improved generalization and robust performance when evaluated on unseen samples compared to the customized CNN and InceptionV3 models, making it suitable for practical sign language recognition applications [13, 28].

2 Literature Survey

Sign language recognition has been an active research area in the field of computer vision and deep learning due to its importance in improving communication for individuals with hearing and deaf individuals [8, 26]. Over the years, various approaches have been proposed using traditional image processing techniques as well as deep learning-based models [25, 27]. Early works in sign language recognition mainly relied on handcrafted features such as edge detection, shape descriptors and skin color segmentation [27, 30]. These methods required careful feature engineering and were highly sensitive to variations in lighting conditions, background noise and hand orientation. As a result, their performance was limited when applied to real-world scenarios [5]. With the technical improvement in deep learning, Convolutional Neural Networks (CNNs) became a more reliable approach for sign language recognition [6, 28]. Several studies have demonstrated that CNN-based models can automatically learn spatial features from hand gesture images, eliminating the need for manual feature extraction [9, 18]. However, many of these approaches used custom-designed CNN architectures, which often suffered from over fitting, especially when trained on limited or imbalanced datasets [6, 29].

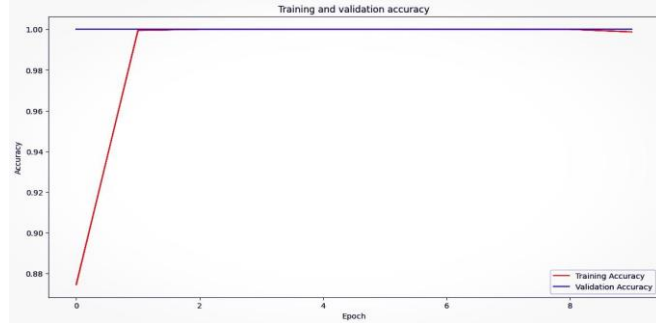


Fig. 2. Illustration of over fitting in CNN model training.

To improve performance, researchers started adopting pre-trained deep learning models such as InceptionV3, VGG, ResNet, and MobileNet using the concept of transfer learning [1, 6 and 8]. InceptionV3, in particular, has been widely used because of its deep architecture and ability to acquire complex features [6, 12]. While these deep models achieve high accuracy, they usually require large computational resources and may still over fit when the dataset contains redundant or highly similar images [3, 9].

Recent studies have shown that lightweight architectures, such as MobileNetV2, are well-suited for image classification tasks where computational efficiency and generalization are important. MobileNetV2 uses depth wise separable convolutions and inverted residual blocks which significantly reduce model complexity while maintaining strong feature extraction capability. This makes the model highly suitable for real-time and resource limited applications [1].

Despite these advancements, many existing works report high accuracy without addressing dataset quality issues, such as duplicate images, having imbalance classes and data leakage between training and validation sets [4, 9]. These issues often lead to misleading performance results that do not generalize well to unseen real-world images [11].

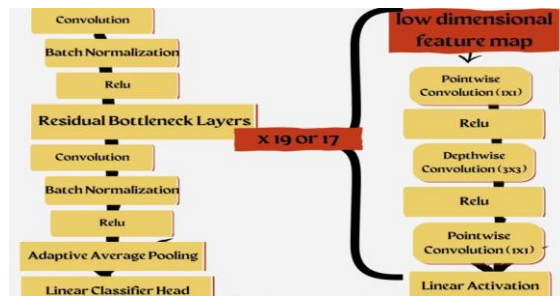


Fig.3. MobileNetV2 architecture with Inverted Residual Blocks

In this work, we build upon existing deep learning approaches by focusing not only on model selection but also on robust data preprocessing, duplicate removal, class balancing using data augmentation, and controlled fine-tuning of a pre-trained MobileNetV2 model. We also perform a comparative analysis with a customized CNN and InceptionV3 to highlight the impact of over fitting and the advantages of using an efficient transfer learning-based architecture [16, 25].

3 Proposed Methodology

The proposed methodology focuses on building a robust and generalized American Sign Language (ASL) recognition system by systematically addressing key challenges such as data redundancy, having imbalance classes and model over fitting. The complete workflow begins with dataset preparation and cleaning, followed by data augmentation, model training using transfer learning, fine tuning and robust validation.

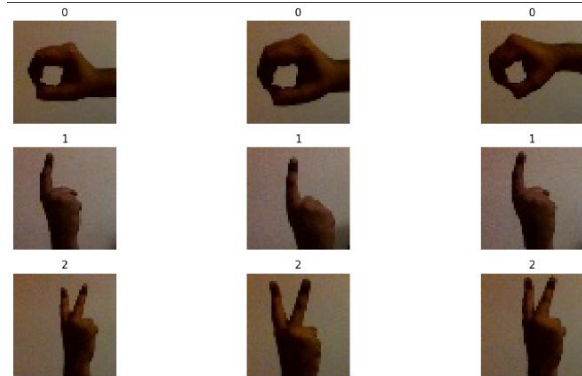


Fig.4. Raw dataset containing duplicates

The American Sign Language dataset used in this study consists of hand gesture images representing digits (0-9) and alphabets (A-Z). During initial experiments it was observed that the dataset contained a significant number of duplicate and visually similar images which resulted in unusually high validation accuracy during early training epochs. This behavior indicated severe over fitting, where the model was memorizing repeated samples rather than learning meaningful gesture representations.

To address this issue a systematic data cleaning procedure was carried out. Initially, corrupted images were identified and removed by checking file integrity using the Python Imaging Library (PIL) and unsupported or invalid file formats were dis-

carded. Duplicate samples were then filtered using perceptual hashing (pHash) implemented through the Image-Hash library. Images that generated identical hash values, corresponding to a Hamming distance of zero, were considered duplicates and only one representative image was preserved for each class. In addition, low-quality samples were excluded by removing images with very small dimensions (less than 40 pixels in height or width) as well as blank or near-blank images. Blank images were detected based on extremely low pixel-intensity variation in gray scale format.

After the cleaning process, noticeable classes with different numbers of images were observed, with certain classes containing significantly fewer samples than others. To mitigate this problem, data augmentation techniques were applied to enhance dataset diversity and balance class distribution. Augmentation operations such as rotating the images in different directions, scaling, translation, zooming and flipping were applied to synthetically generate new training samples. Through this process each class was expanded to contain at least 1000 images. This step improved the model's robustness to variations in hand orientation, scale and environmental conditions.



Fig.5. Real and augmented validation images

Despite these improvements over fitting persisted when validation data was derived solely from the same dataset distribution. To further enhance generalization a custom validation dataset was created using independently captured hand gesture images obtained under different devices, lighting conditions and backgrounds. These images were further augmented to simulate realistic variations. This validation strategy inspired by existing research ensured evaluation on out-of-distribution samples and encouraged the model to learn generalized gesture representations rather than dataset-specific patterns. For feature extraction and classification, MobileNetV2 was selected due to its lightweight architecture and efficiency. The network was initialized using pre-trained weights from Image Net to facilitate transfer learning. Allowing reuse of learned low-level and middle level visual features.

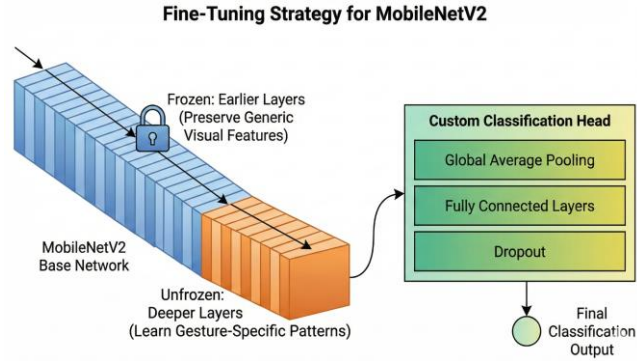


Fig. 6. Fine-Tuning of MobileNetV2

A fine-tuning strategy was applied by freezing early layers and selectively unfreezing deeper layers to learn gesture-specific patterns. The base network was extended with a custom classification module that includes global average pooling, dense layers, and dropout for regularization to improve generalization and reduce over fitting.

4 Results and Discussion

The customized CNN was intentionally designed as a shallow baseline to reflect commonly adopted initial architectures in gesture recognition and educational image classification tasks. In many practical scenarios, simple CNN models are often assumed to be sufficient when high training accuracy is observed. By including this baseline, the objective was to highlight the limitations of shallow CNN architectures when applied to a complex 36-class ASL recognition problem. Although the customized CNN achieved high training accuracy, it failed to generalize effectively, clearly demonstrating severe over fitting.

The comparison was therefore not intended to position the customized CNN as a competitive alternative to MobileNetV2 but rather to emphasize the necessity of deeper architectures and transfer learning for robust ASL recognition. The predictable performance gap reinforces the importance of structured feature extraction and fine-tuning of pre-trained models. This experimental contrast helps dispel the misconception that shallow CNNs are adequate for real-world gesture recognition tasks and motivates the adoption of lightweight yet powerful architectures such as MobileNetV2.

Table1. Performance comparison of different models on the ASL dataset

Model	Training Accuracy	Validation Accuracy	Test Accuracy
Customized CNN	99	99.5	78
InceptionV3	99.8	99	81.6
MobileNetV2(proposed)	97	95	94

From the above table we got the observation that model achieves high accuracy with internal validation but compromise with performance on external images. While getting high training accuracy but fails to generalize unseen samples effectively and MobileNetV2 provide stable generalization with reduce over fitting ,

Table 2. Impact of data cleaning on MobileNetV2 Performance

Dataset version	Training Accuracy	Validation Accuracy
Customized CNN	99	99.5
InceptionV3	99.8	99
MobileNetV2(proposed)	97	95

Due to duplicated values severe over fitting and generalization improves with cleaned dataset. To analyze the effectiveness of the data cleaning stage, an ablation study was conducted using the proposed MobileNetV2 model. When trained on the raw dataset containing duplicate and visually similar samples, the model achieved unrealistically high validation accuracy in the initial epochs, indicating severe over fitting. After removing duplicates and low-quality images, the validation accuracy followed a gradual and stable learning trend. This confirms that data cleaning plays a crucial role in improving model generalization and preventing memorization of repeated samples.

The effectiveness of the proposed ASL recognition system was evaluated using training and testing datasets. Training was performed using augmented data generators while regularization techniques such as adding dropout layers, early stopping logic and adaptive learning rate scheduling were employed to stabilize learning and prevent over fitting.

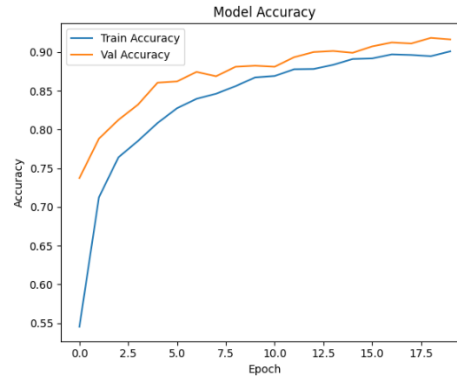


Fig.7. Training and validation accuracy of the optimized model

The model demonstrated steady convergence, with validation accuracy improving gradually across epochs rather than exhibiting unrealistically high accuracy during early stages. This behavior confirmed that the applied data cleaning and validation strategies successfully reduced over fitting. Evaluation on the independent validation dataset showed strong generalization performance highlighting the model's ability to recognize gestures captured in unseen environments.



Fig 8. Sample of accurate real-time prediction

The final trained model achieved strong performance in terms of classification accuracy on the test dataset outperforming baseline models trained without extensive data cleaning and validation separation. Qualitative analysis further validated model performance as shown through correct predictions on real-world hand gesture images collected independently.

Comparative experiments conducted using a customized CNN and InceptionV3 resulted in higher training accuracy but suffered from severe over fitting. In contrast, the MobileNetV2-based model achieved a better balance between accuracy and generalization, making it more suitable for real-world deployment scenarios.

5 Conclusion

In this research, an efficient American Sign Language recognition system was developed using MobileNetV2 with transfer learning and fine-tuning to classify hand gestures representing digits (0-9) and alphabets (A-Z). The work highlights the critical role of robust data preparation, where extensive dataset cleaning, duplicate removal, class balancing through data augmentation and the creation of an independent validation dataset were carried out to reduce over fitting and improve model generalization. A comparative analysis with a customized CNN and InceptionV3 revealed that, despite achieving high training and validation accuracy, these models struggled to generalize to unseen data due to dataset similarity and over fitting. In contrast, the proposed MobileNetV2-based approach showed stable and reliable performance on external images, indicating effective pattern learning. The results confirm that combining transfer learning with proper dataset management and validation strategies leads to a more accurate and practically deployable sign language recognition system. This work also emphasizes that model accuracy alone is not a sufficient indicator of real-world performance. Future work may focus on real-time implementation, mobile deployment and training with larger, more diverse datasets to further enhance system robustness.

6 Future Work

Although the proposed system demonstrates improved generalization for static sign language image classification, several extensions can be explored in future work to enhance its real-world applicability. One major direction is the transition from image-based recognition to video-based sign language understanding, where temporal information plays a crucial role. Transformer-based models such as Mobile BERT can be investigated to model sequential dependencies and improve contextual understanding in continuous sign language recognition. In addition Media Pipe Holistic can be integrated for efficient hand, pose and facial landmark detection from video streams. Landmark-based representations may reduce background dependency and improve robustness across different environments and users. The extracted landmarks can be used as structured inputs to transformer or hybrid deep learning models for improved performance.

Furthermore, the system can be extended to support real-time text and speech output by integrating text-to-speech frameworks such as Google Text-to-Speech (TTS). This enhancement would enable end-to-end sign to text and speech translation, making the system more practical for assistive communication applications. Future

work may also include mobile deployment, optimization for low-latency inference and evaluation on larger more diverse sign language datasets.

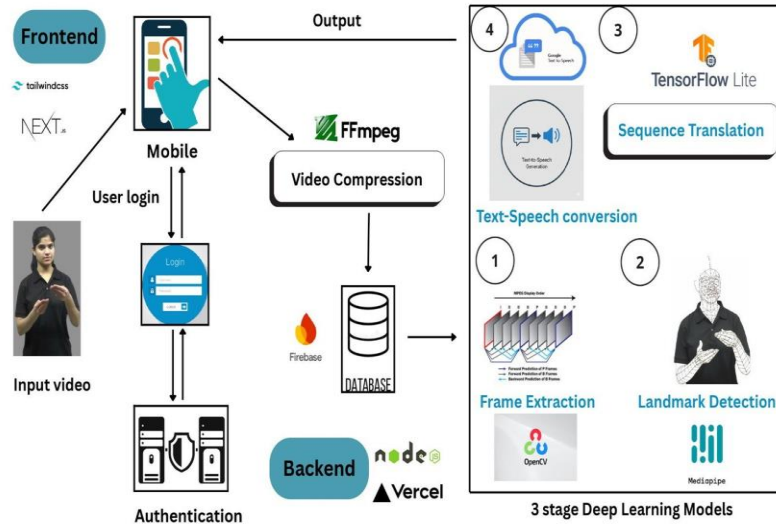


Fig. 9. Future System design for real-time sign language translation

The proposed system implements an end-to-end pipeline for translating sign language video inputs into text and voice through a structured frontend, backend architecture. The process begins at the frontend, where users securely log in and record or upload sign language videos using a mobile interface developed with Next.js and Tailwind CSS. To reduce transmission and storage overhead, the uploaded video is compressed using FFmpeg and then stored in a Firebase database. The backend retrieves the stored video then extracts individual frames using Open CV, preparing the visual data for further analysis.

The extracted frames are processed using Media Pipe Holistic to detect hand, pose, and facial landmarks, which transform raw visual input into structured representations that are less sensitive to background variations. These landmark sequences are then passed to a Tensor Flow Lite based deep learning model that performs sequence translation to generate meaningful textual output. Finally, the identified text is converted into natural speech using Google Text-to-Speech and both text and audio outputs are delivered back to the frontend. This modular design enables efficient processing, improved generalization and real-time sign-to-speech translation suitable for practical assistive communication systems.

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14

31.